

AR Music Visualizers: Application Space and Design Guidelines

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Figure 1: Examples from our design idea generation study that aims to bring out different benefits of AR music visualizer: (a) Offer a new form of entertainment; (b) Facilitate the process and enhance creativity for music and audiovisual experience production; (c) Enhance the effect of music on emotion regulation; (d) Heighten awareness of the real world; (e) Create a shared or social musical experience.

ABSTRACT

Music visualization has recently become a popular application of augmented-reality (AR) technology. However, despite their various designs and functionalities, their potential benefits and the effectiveness of their designs are not well explored. By interviewing experts and generating and evaluating design ideas, we offer two contributions: (1) the application space of AR music visualizers, defined by the categories of their potential benefits and the constraints from their contexts of usage; and (2) a four-step design guidelines to design a visualizer that achieves the intended benefit(s), while appropriately handling the contextual constraints. We believe that this paper can provide a concrete basis for designing effective AR music visualizers in the future.

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CCS CONCEPTS

• **Human-centered computing** → **Visualization design and evaluation methods**; *Ubiquitous and mobile computing design and evaluation methods*; Interaction design process and methods.

KEYWORDS

augmented reality, music visualizer, design guidelines

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1 INTRODUCTION

Imagine that you are reading a book and listening to soft instrumental music, which you know can increase your focus. You have a music visualizer which displays a plain color in the periphery that fits the music's emotion over your surroundings, blocking out the distractions, and also ripples to the music's rhythm to keep the atmosphere from being too static and dull. Such systems can be implemented as an augmented-reality (AR) application, which

allows you to remain aware of the real world while enjoying the benefits of the add-on visualization.

AR music visualizers are becoming popular. Their designs vary in the components of music they respond to as well as the visual effects they generate. For instance, Beatsy detects a real-world surface and creates an outward ripple effect according to the amplitude [1], while MotionAR has several floating balls, each activated by different frequency ranges [16]. AR technology has the potential to connect listeners to the music more strongly through embodied cognition [14], enhance the music's effects on emotion regulation [3], and offer a new realm of possibilities of music listening experience in everyday life.

Although many AR music visualizers available now have the aforementioned potentials, their benefits and application space remain unexplored. Taking the example above, we do not know the demographic of listeners who would benefit from such applications, the tasks they could be doing while listening to music (e.g., reading a book), or the possible benefits of the music and visualization (e.g., increase the focus and maintain alertness). Most importantly, we do not have the framework on how to design an AR music visualizer to suit different combinations of these variables (e.g., what should a visualizer do to increase the focus on novel readers?) Without this knowledge, we cannot effectively design AR music visualizers to maximize their benefits.

In this study, we aim to answer the following research questions:

- **RQ1.** What is the application space of AR music visualizers?
- **RQ2.** How can we design an AR music visualizer to maximize its benefit in a particular usage context?

We interviewed AR designers, developers, and researchers, and generated design ideas of AR music visualizers to explore what makes a design effective. Consequently, we offer two contributions:

- **C1.** An application space of AR music visualizers, defined by two variables: the potential benefit of the visualizers and the constraints imposed by their context of usage.
- **C2.** Four design guidelines for designing an effective AR music visualizer.

2 RELATED WORK

2.1 Design space of AR applications and visualization

There have been several studies that explore the design space of augmented reality, focusing on specific device/installation, interaction techniques, visualization style, and context of usage, and so on. AR music visualizers fall into the conceptual space of pervasive AR applications, which are context-aware and context-driven [6]. This informs us that the application space of AR music visualizers will likely be defined by their contexts of usage, and that, beyond the effectiveness of the application, the design should also cover safety consideration for the users [12].

Visualization is a popular application for AR technology. Zollmann et al. proposed a taxonomy of AR visualization, which is defined by visual attributes of the visualization design, such as visibility and depth cues [17]. Lindbauer et al. introduced a design space for projection-based AR visualization with the purpose of manipulating the user's perception of reality. The design space

covers possible visual effects and how they could be implemented [9]. Mori et al. conducted a survey on the techniques, usages, and devices for diminished reality, where the AR technology visually removes objects from reality instead of adding to the reality [11]. Our project will build upon all these framework by applying them specifically to music visualization tasks.

2.2 AR Music Visualizers

There are many examples of AR music visualizers both as design experiment projects and commercially deployed applications. Most applications are designed for smartphone or tablets - we found only one application for wearable devices [13]. Most applications do not have any features other than displaying the visualization, and *Fragmented Reality* is the only application for facilitating music composition, production, and collaborative performance [7].

The visualization effects can be divided into three categories. The first type, including *Music MotionAR* [16] and *Spotify AR visualizer* [4], directly displays visual objects without establishing any relationship or interaction between the visuals and the real world. The second type has the visual objects interact with the real world. For example, *AR Audio Spectrum 3D* places the visualization on a real-world surface and cast shadow of the visual object onto the surface [2]; *Music AR* makes balls bounce on the floor, walls, and ceiling of a room. The last type manipulates real-world surfaces by . The third type manipulates the real world surfaces and objects by replication or distortion in response to music. Some examples of this type are *Beatsy* [1], *MusicAR* [10], and the *Markerless visualizer* [15].

In this project, we aim to explore the potential benefits and visualization designs of AR music visualizers, covering and going beyond the existing applications.

3 METHOD

We conducted two studies simultaneously: expert interview and design ideas generation. We triangulated the data from both studies to generate the findings.

3.1 Expert Interview

During literature reviews, we curated a list of experts in the fields related to AR music visualizers. We recruited four experts (all male). P1 and P2 are from industry and have deployed commercial AR Music visualizers. P3 and P4 are from academia and have designed music-related, immersive systems.

We interviewed the experts via Zoom video call for approximately 45 minutes, during which we asked for their opinions about potential benefits and limitations of AR music visualizers, and asked them to describe their design considerations and processes for immersive music technology projects they have worked on. The participants received CAD80 as compensation. We recorded the interviews and re-watched them to take notes of important ideas.

3.2 Design Ideas Generation

We brainstormed 30 design ideas for AR music visualizers, focusing on the contexts where users listen to music in the background while doing other primary tasks. We created low-fidelity design sketches in Microsoft PowerPoint (examples shown in Figure 1 and

2.) We asked 7 graduate students (P5-P11; 1 female and 6 male) with experience in human-centered design or immersive technology development to subjectively evaluate the designs and give comments on why they think some designs would work better than others. We carried out the evaluation and the follow-up discussion via asynchronous messaging with the participants.

4 FINDINGS

4.1 Application Space

AR music visualizers have the potential to benefit all kinds of musical activities, including composing, producing, performing, consuming, and sharing. However, to define the application space of AR music visualizers, we need to specify what benefits visualization in augmented reality can bring to these musical activities and in what contexts. Based on our study, we presented five categories of potential benefits of AR music visualizers and three variables that contribute to the contextual constraints of their usage.

4.1.1 Benefits of AR music visualizers. It is unclear what the benefits or purposes of existing AR music visualizers are. However, the expert participants believe that AR music visualizers have many potential, tangible benefits, which lie in the intersection between the benefits of visualization in augmented reality and the benefits of music visualization. The benefits fall into five categories, as follow:

- **Offer a new form of entertainment.** Both developers of commercial AR music visualizers (P1 and P2) said their applications were meant to be "a toy." P1 said "[their AR music visualizer] is a good way to capture something cool and place it on a fun surface" and that "it just makes the experience of the music a little bit more engaging." P2 mentioned that the visualizer could be something to help promote music sales: "I think bands can use AR. And of course, studios. To sell an additional experience when you make [a new album]... you download the app, you get all the music, plus you get the interactive video, where you can make the band play on your kitchen table with AR."
- **Facilitate the process of and enhance creativity for music and audiovisual experience production.** AR music visualizers have a strong potential to benefit creative activities, such as music composition, music production, and audiovisual experience production. Music performance would not be confined to the instrument and open up new possibilities of performances. For instance, P2's application was developed under the vision that "music performance would not be confined to the instrument." Similarly, P4 said that it would be beneficial to "use the technology as an enabling power to get people without having to have had the years of training, nonetheless, to do musically, sonically interesting and sophisticated things to the music itself." P4 also mentioned a different potential benefit of an immersive visualizer for "experimenting with different technologies for offering new audiovisual content and creating workflows for creative artists to make immersive artworks."
- **Enhance the effect of music on emotion regulation.** Listening to music helps regulate emotions. When the visualization interacts with or emphasizes certain aspects of the

music, it could create the synesthesia effect that transforms the experience as a whole. Based on the design idea generation study, 5 out of 7 participants commented without any prompts that they thought some of the designs can help regulate emotions in many everyday life contexts that were not possible with 2D visualization on computer monitor or with virtual reality. For example, P6 commented that "I would love [idea in Figure 2 (left)] to come true for the boring chores I do." P10 commented "I think [idea in Figure 2 (middle)] would be extremely useful in helping increase focus." And P11 commented that "[idea in Figure 2 (right)] might make me more alert while driving a long distance."

- **Heighten awareness of the real world.** When people listen to music with their headphones, they may become less aware of the real world. P3 mentioned that the visualizers would be good at "bringing awareness to the power of sound, to show any impact of sound, visually ... [For example], the difference of loudness: what does that do? How might it affect your perception? How it might be damaging your ears?" P2 suggested that visualization in augmented reality can help direct their attention to events in the real world. For example, in a hypothetical situation where the user listens to music while driving a car, "[the visualizer can make] the music focuses on the oncoming car or the pedestrians to enhance them. So, you can argue that the visualizations are enhancing your ability to perceive." Although such visualization does not necessarily have to interact with music, their interaction would make experience more seamless, as though the music in the headphone is embodied as a part of the real world.
- **Create a shared or social musical experience.** AR music visualizers can situate music at specific locations in the real world and allow for applications like collaborative performance or location-based music recommendation. One application that P4 mentioned was "network music performance, where we've connected two stages in two remote countries with two stages, two performers on each stage, collaborating over the network." The visualizer could "project the presence of the remote musician as an augmented reality layer." P1 suggested that "you can use these codes, that you can either print out or have in little stickers. And if you put those on the wall, you can use an iPhone to go up and scan the code and actually launch the visualizer around it. So, you can have this discoverable experience out in the real world ... They have to physically be there to be able to scan these stickers, and then they could meet other people, and that could be part of the experience."

4.1.2 Contextual constraints. Because AR visuals are overlaid onto the real world, the contexts in which the applications are used present constraints on the design of the visualizers. The context of usage encompasses the characteristics of the environment, the activities or tasks that the users may be performing, and how the users will be using the visualizers. Based on our study, we found that the contextual constraints of AR music visualizers can be largely described by three variables: visual load, cognitive load, and device.

- **Visual load** refers to the amount and characteristics of visual information in the context of usage. AR music visualizers



Figure 2: Examples of AR music visualizer design sketches: (left) a dancing line around the sink for people who listen to music while doing the dishes; (middle) blurred surrounding around a book with modified color tone for people who listen to music while reading; (right) a single volume bar for people who listen to music while driving.

present additional visual information on top of the visual load of the context, so they are most suitable for contexts with low visual load, i.e., have low information density, repetitive information, or visual information that coincides with the purpose of the AR music visualizers. On the other hand, they are not suitable for contexts with high visual load, i.e., have high information density or have sensitive information, unrelated to the purpose of the AR music visualizers, that requires apt attentions. Some examples of unsuitable contexts are crowded places and situations with safety risks (P5, P7, P10).

- **Cognitive load** refers to the intensity of cognitive processing that the user spends in the context of usage without listening to music. This variable is only important when the user performs a primary task that is not related to music while listening to music and using the visualizer as a secondary task in the background. AR music visualizers are more suitable when the user experiences low cognitive load, such as when their primary task only requires unconscious processing, and are less suitable for tasks when the user experiences high cognitive load, such as when they are learning or exercising higher reasoning. It should be noted that this variable depends on both the individual user and the nature of the primary tasks. For example, some participants said that they would like to use AR music visualizers while writing (P11), whereas some say they need total concentration while writing and would not even listen to music while doing it (P8, P9).
- **Devices** refer to the technology for displaying the AR visualization. According to P1-P4, the limitation of current device technology is the biggest obstacles in designing and deploying AR music visualizers. The most prevalent device for AR applications are handheld devices, such as smartphones and tablets. However, these devices require the user to have at least one free hand to hold them up and require the user to focus on the screen to see the visualization. P1 explained the problem in a hypothetical concert scenario: "[if] you want to have music visualization going on, and if you take out your phone to look around, it takes you out of the moment." P3 brought up another problem in a relaxation scenario: "if you're asking people to hold up a device for a certain period of time to chill out, at a certain point, my hand will get tired - I'm no longer chilling." Therefore, they are not suitable

for contexts where users experience encumbrance or wider field of vision. Other devices, such as head-mounted displays (HMD) and AR glasses, though allow free-hand usage and wider perspectives, may impede users in the contexts that require high mobility, such as dancing or running. P1 also mentioned that they were still "not good enough to render immersive visuals [that manipulate real-world surfaces]", though this limitation is likely to disappear in the future.

4.2 Design guidelines

Based on the expert participants' descriptions of their design processes and the comments on our design idea sketches, we generated a four-step design guidelines for creating an effective AR music visualizer.

- (1) **Specify the purpose of the visualizer.** AR music visualizers should strive to have a clear benefit instead of being general-purpose. A category of benefits of Section 4.1.1 can describe a high-level purpose of a visualizer, but the designer should narrow it down as much as possible. For instance, rather than setting the purpose to be enhancing emotion regulation, it could be narrowed to enhancing the user's focus on their primary task while they listen to music in the background; rather than enhancing creativity, it could be narrowed to generating composition ideas based on the user's surrounding. The purpose of the visualizer will inform the effective visualization design and the features the visualizers must have.
- (2) **Specify the context of usage.** As described in Section 4.1.2, the context of usage can present many limitations to the application and consequently inform the visualization design. Therefore, the designer should be specific about when, where, and how the users will be using the application. As an example, for the purpose of enhancing focus mentioned above, its context usage could be for users who read while listening to music. This context would tell us that the user will stay still for a long time, but their hands will not be available, since they have to hold a book or an ebook reader. It also informs us that the necessarily visual information will be in the center of their field of vision, and that the cognitive processing will likely be medium to high. Similarly, for creativity purpose, the context could be for an improvisation performance on a piano in an auditorium, where the user

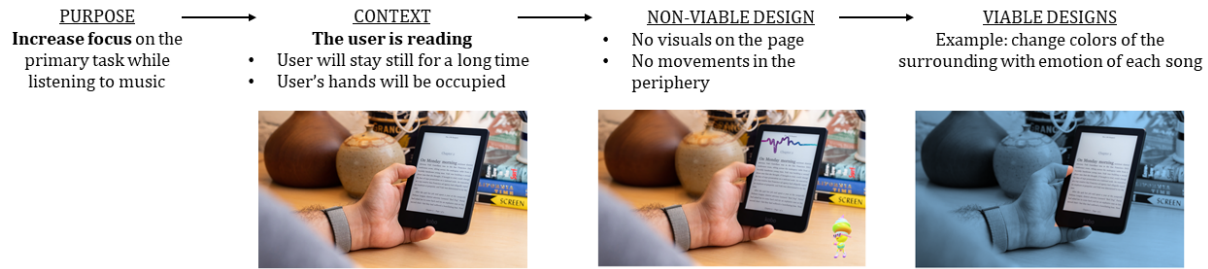


Figure 3: Example of how the proposed design guidelines can be utilized to design an AR music visualizer.

will need to be aware of the stage but not that much of the audience, and both of their hands will be occupied.

- (3) **Don't let the visualization occlude or interact with things that are crucial to the context.** Many of the generated design ideas did not work are because they distracted or hindered the user from performing on a task, whether the task is related to music or not. The applications need to identify if there are any crucial areas in the field of vision that the visualization cannot interact with, such as overlaying graphics on the book the user is reading or occlude the instrument or sheet music stand while the user is playing. It should be noted that these crucial components do not only depend on the nature of the task but also where and when the task is performed. For example, if a user plays with an AR music visualizer as their primary task in a crowded shopping area, the visualization should not overlay on other people that might walk into them; if a user uses it for a music performance on stage, it should not manipulate the edge of the stage, risking the musician's safety. These aspects should be identified when defining the context of usage in step 2.

- (4) **Design visualization that interacts with the real world.** Anything that are not crucial to the context of usage can be safely utilized to achieve the purpose of the visualizers. For example, when a user is reading, the color of their surrounding and very small movements in their periphery may not be crucial to the task, so they can be manipulated. Moreover, all participants say that it is better to have the visualization interact with the real world, instead of simply overlaying irrelevant graphics on them. For example, when using a ripple visual effect, make real-world surfaces ripple rather than visualize virtual objects that ripple.

5 DISCUSSION

5.1 Connection to existing AR design guidelines

The steps in the proposed design guidelines in Section 4.2 are subsets of AR application design principles proposed by Ko et al, which consist of five principles and focus of application usability [8]. Because AR music visualizers focus on more on visualization and less on interactivity, we emphasize the importance of context-based visual design in step 3 and 4. Our guidelines also provide a sequential order of actions that tell designers what attribute of the project to

specify first and how a step connects to the next, making it easy for designers to follow.

Our design guidelines are high-level; that is, it does not describe the details of visualization attributes or interface designs. Therefore, it should be used in conjunction with other design guidelines or design space that are specific to the context of usage, such as the guidelines for designing mobile AR applications [5], the design space of AR data visualization [17], and the design space of augmented reality for manipulating visual perceptions [9].

5.2 Applying guidelines to existing AR music visualizers

Fragmented Reality has the purpose of utilizing fractals concept to facilitate composition and performance [7]. Therefore, its visualization design is motivated by music theory. It is not designed for a specific context of usage, group of target users, or location, so its design may present some safety or efficiency issues. For example, if a producer uses it in a music studio with a lot of machines and instruments, the visualization may occlude some of the real world objects. This emphasizes the importance of step 2 and 3 in the proposed design guidelines.

Apart from Fragmented Reality, other existing AR music visualizers do not provide state any specific purposes, benefits, or contexts of usage. Based on their demo videos, we concluded that their benefits are likely to be toys - offering a new form of entertainment. These visualizers are likely motivated by the designer's desire to play with and test the boundary of AR technology rather than specific benefits of the applications. Therefore, we concluded that their design processes likely did not follow our proposed design guidelines and may present safety issues if the users use them in contexts with high visual load.

It should be noted that these visualization designs allow varying levels of customization so that the users can optimize their entertainments. However, we were not able to deduce a guideline on optimizing the balance between customizability and automated adjustment of the visualization based on a visualizer's purpose and use context. This is a potential issue for future work.

5.3 Effects of the AR technology

The contextual constraints presented by the limitation of the device will likely change as the hardware technology rapidly progresses. To our knowledge, there is currently no music visualizer designed specifically for AR glasses, since it is less accessible by consumers

and lacking in resources for designers and developers. Even our expert participants have no experiences working with them, so they are not confident to discuss the characteristics of AR glasses. As the hardware technology develops, our findings on the application space of AR music visualizers will need to be expanded, correspondingly.

On top of the hardware technology, the development of computer vision techniques also affects the application space of AR music visualizers, since it defines the extent to which the visuals in augmented reality can interact with the real world. The recent advancement in lightweight, surface detection algorithms enable applications like Beatsy [1], Markerless [15], and Music AR [10]. In the future, if the devices are able to detect even more comprehensive characteristics of the environment, the application space of AR music can be expanded as well.

5.4 Future work

Our work can benefit from a larger pool of expert participants, since our interview data is very diverse, implying there could be more design considerations that our participants did not cover due to their specific field of expertise. We would also like to demonstrate and evaluate the proposed design guidelines by using it to design, implement, and conduct user studies on AR music visualizers for different categories of benefits and different contexts of usage.

6 CONCLUSION

In this paper, we conducted a two-part study to explore the potential benefits and limitations of AR music visualizers. As a result, we proposed an application space of AR music visualizers, defined by categories of benefits of such applications and their contextual constraints. We also proposed a four-step design guidelines, which utilize the dimensions of the application space to help design an AR music visualizer that is effective for specific purpose and context of usage. Although our findings greatly depend on the current state of technology, we believe that this paper can be a concrete basis upon which to improve and expand in the future.

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